Caetano Marcio Monderen Milera

Student

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EDUCATION

09/2026 – 09/2027
Belgium
Postgraduate: AI for the common good
Vrije Universiteit Brussel

09/2025 – 09/2026
Brussels, Belgium
Master Applied Informatics: Artificial Intelligence
Vrije Universiteit Brussel

09/2021 – 08/2025
Brussels, Belgium
Bachelor Artificial Intelligence
Vrije Universiteit Brussel

01/2018 – 03/2018
Tarrytown, New York, United States
Exchange trimester
EF
EF (Education First exchange for academic English)

Mechelen, Belgium
High School Science and Mathematics
Sint Rombouts College

NOTABLE PROJECTS

Master's Thesis: Modelling Melodic Similarity
Skills: Signal Processing, Music Information Retrieval, Mathematical Modelling, Python
- Investigated how to represent and compare melodic phrases in the Discrete Resonance Space. Developed a pipeline that transforms melodic contours into resonance representations and compares them using cosine similarity and Hilbert-space distance. The strongest approach combined Hilbert-space distance with a Gaussian RBF. Additional experiments examined sensitivity to melodic changes and invariance to pitch transposition.

VRT AI avatar
Skills: Prompt engineering, SQL database handling, LLMs
Development of the early stages of an AI avatar using HeyGen, databases and ChatGPT-4. Further improvements include the incorporation of NVIDIA ACE and Mistral as the main LLM.

Wildfire Resource Allocation: Reinforcement Learning
Skills: Reinforcement Learning, Markov Decision Processes, Reward Design, Decision Support
- Designed a reinforcement learning framework for coordinating aerial and ground firefighting resources during active wildfires. Formulated the dispatch problem as a Markov Decision Process, defining observations, actions and rewards that account for fire spread, resource availability, human safety and environmental damage. Developed a simulation-based training and evaluation proposal, examining operational constraints, feasibility and the challenges of transferring learned policies to real emergencies.

Secure Messaging Application: Security in Computing
Skills: Electron, Node.js, SQLite, Web Security, Cryptography
- Strengthened an initially insecure chat application by implementing user authentication, bcrypt password hashing, persistent storage and protection against SQL injection and cross-site scripting. Added client-side message encryption using AES-GCM, RSA-OAEP room-key distribution and TLS transport encryption. Implemented membership-based access controls and room-key rotation to prevent departing channel members from decrypting future messages.

Traffic Prediction with Apache Spark: Big Data Processing
Skills: Apache Spark, Spark ML, Structured Streaming, Apache Kafka, Feature Engineering

- Built a traffic prediction pipeline combining distributed batch processing with real-time streaming. Engineered temporal features from traffic speed and volume measurements, then trained Random Forest models to predict speeds between 5 and 30 minutes ahead. Integrated Kafka and Spark Structured Streaming to generate live five-minute forecasts, using per-road historical state, watermarking and checkpointing.

Bachelor's Thesis: Motor Imagery EEG Classification
Skills: Deep Learning, EEG Signal Processing, Convolutional Neural Networks, Brain–Computer Interfaces

- Investigated deep learning methods for recognising imagined movements from EEG recordings. Compared EEGNet, ShallowNet and DeepNet on the BCI Competition IV Dataset 2a, using 22-channel recordings to distinguish four motor imagery classes. Examined the suitability of different convolutional neural network architectures for brain-signal classification.

Agend Based System for Grid Packing
Skills: Agent Based Systems, Statistics , Python(pandas, numpy, seaborn, . . .)

- Engineered an Agent-based system to simulate an optimised organisation grid where robots(agents) hover across a grid to sort, place and retrieve a list of items. This model was made for real-life applications where warehouses / industrial storage rooms and supermarkets can be fully automated. Agents communicate with each other and use heuristics for optimised pathfinding.

Euthopia Pacman Capture the flag tournament
Skills: Heuristics, python .

- The tournament consists of a multiplayer capture-the-flag variant of Pacman, where agents control Pacman and ghosts in coordinated team-based strategies. Students from different EUTOPIA universities compete with each other through their programmed agents.

SKILLS

Machine Learning ●●●●○
TensorFlow, PyTorch, Scikit-learn, . . .

Neural Networks ●●●●○
CNN / RNN / DNN

Programming ●●●●○
Python / OOP / Prolog /

Data Analysis ●●●●○
Pandas, NumPy, Matplotlib, ...

Reinforcement Learning ●●●●○
Markov Decision Processes, Reward Design, Sequential Decision-Making

AI Ethics & Governance ●●●●●

Big Data & Stream Processing ●●●○○
Apache Spark, Spark ML, Apache Kafka, Structured Streaming

Advanced Natural Language Processing

Signal Processing ●●●●○
EEG Analysis, Fourier Analysis, Spectral Representations, Melodic Similarity

Cybersecurity
Authentication, End-to-End Encryption, SQL Injection Prevention, XSS Protection, TLS

CERTIFICATES

TOEFL English proficiency
C1 CEFR

Bachelor Artificial Intelligence
With distinction

Master of Science in Applied Informatics: Artificial Intelligence
With great distinction

LANGUAGES

English ●●●●○
TOEFL

Spanish ●●●●●
Mother tongue

Dutch ●●●●●
Mother tongue

French ●●○○○

ORGANIZATIONS

03/2025 – Present

Informatica Groep
Board Member
Informatica Groep is a student association of the faculty of computer science and AI. We maintain servers, computer rooms and work on events like capture the flag and students to industry, where we make the bridge between students and corporations.

08/2021 – present
Boortmeerbeek, Belgium

Chiro Boortmeerbeek
Head Leader
I serve as a group leader and head leader in a youth scouting movement, managing both administrative tasks and leadership responsibilities for a 5th consecutive year.

2018 – 2020
Lima, Peru

SSCC Belen - Misiones
Volunteering action to aid the needs of the Amazonian tribes, such as healthcare, education, sustainable development, and cultural preservation.

SOFT SKILLS

Time management

Adaptability

Interpersonal skills

Teamwork

Communication

Leadership

NOTABLE COURSES TAKEN

- Foundations of language
- Roots of the AI
- Human-AI interaction
- Cognitive modelling
- Agent-based Systems
- Multimedia Processing tools
- Advanced Natural Language Processing
- Cybersecurity
- Cognitive Psychology lecture
- Machine Learning
- Discrete Mathematics
- Algorithms and Data Structures (1 & 2)
- Bayesian statistics
- Calculus (1 & 2) and linear algebra
- Reinforcement Learning